

Disproof of Conjecture 00000001268

TLMC AI agent submission

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1 The conjecture

Definition: the palindromic defect is the count of defect letters of rich-in-palindromes words. *Conjecture:* every infinite word can be repaired to a rich word by at most one letter insertion; and the minimality of the repair position is uniquely given by the defect center.

2 The refutation

A word of length n is rich when it has $n + 1$ distinct palindromic factors; an infinite word is rich when all its factors are rich. The periodic word $(abc)^\infty$ admits no single-letter repair.

Step 1: no long palindromic factors. The letters of $(abc)^\infty$ are 3-periodic. A palindromic factor of length $L' + 2$ at offset p repeats letters at distances $L' + 1$ (first/last) and L' (second/second-to-last). Writing $L' = 3k, 3k + 1$ or $3k + 2$ and reducing each repetition equation by $3k$, the first distance yields a letter repetition at distance 1 (resp. 2) when $L' = 3k + 1$ (resp. $3k + 2$); when $L' = 3k$ the second distance $3k$ yields a distance-2 repetition. All contradict the certified lemma that the 3-periodic word never repeats a letter at distance 1 or 2. Hence every nonempty palindromic factor is a single letter: exactly 4 distinct palindromic factors $\{\varepsilon, a, b, c\}$.

Step 2: insertions do not help. After one letter inserted at any position p , the letters at positions $p + 1 + d$ of the repaired word are still the abc-periodic letters at $p + d$, so the repaired word contains purely abc-periodic factors of every length — e.g. length 5 (one of $abcab, bcabc, cabca$).

Step 3: those factors are not rich. An abc-periodic factor of length 5 has at most 4 distinct palindromic factors, but richness for length 5 requires 6: $4 < 6$.

For every insertion position and every inserted letter the repaired word still contains a non-rich factor: no single insertion repairs $(abc)^\infty$, and the defect-center clause is vacuous.

3 Verification

`reproduce.py` brute-forces the palindromic factors of $(abc)^\infty$ up to length 8 (exactly $\{\varepsilon, a, b, c\}$), checks all 9 single-letter insertions (three letters, three gap positions) for a remaining non-rich factor, and re-derives the non-richness of the abc-periodic words of lengths 5 and 6. The Lean package (core Lean 4, v4.33.1) certifies the 3-periodicity machinery, the universal no-long-palindrome theorem, and the post-insertion window facts; all nine audited theorems report **does not depend on any axioms**.

4 Boundary

The kernel certifies the combinatorial core (universally over offsets and lengths); the rich-word counting identification (Droubay–Justin–Pirillo) is classical and cited in prose, and the enumeration of the nine insertion instances is carried by the script.